Lab 3: GenAI for HRI

Evaluating Camera-Informed Adaptive Guidance in Human–Robot Mystery-Solving
INFO 5356: Introduction to Human-Robot Interaction

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1. Introduction

Social robots in museums, learning settings, and other collaborative tasks must decide *when* help is useful. An unsolicited hint can interrupt a person who is reasoning independently and make the robot feel intrusive, whereas help that comes too late can leave the person stuck and frustrated. Timing guidance well is therefore central to HRI research on socially acceptable robot assistance. Vision-language models (VLMs) make it possible for a robot to use a camera frame, a typed answer, and task context when deciding how to respond, but visible cues such as posture or facial expression do not reliably reveal a person's internal state. We study this problem with Reachy Mini in a structured mystery-solving task, focusing on a VLM-based guidance module inside an end-to-end robot system rather than on the VLM's predictions in isolation.

Prior work shows that confusion in situated, task-oriented HRI has multimodal behavioral correlates, but that observable behavior is not definitive evidence of what a person is thinking [1], [2]. Recent work on generative AI in social robots has examined design considerations for LLM-powered HRI [3], vision-enabled dialogue [4], user-enjoyment prediction [5], and multimodal collaborative storytelling, where turn-based interaction and inference delay were limitations [6].

The gap is that these studies do not directly compare, in a controlled reasoning task with identical hints, a robot that always offers help with one that decides when to offer help from a visual cue and typed evidence. Such a comparison matters because adaptive help is often assumed to be better, while it may also be perceived as arbitrary, or simply reduce the number of useful offers.

We address this gap with a within-participant comparison of two guidance strategies. In *fixed-point* guidance, Reachy asks "Would a hint help?" at every checkpoint without using the camera. In *camera-informed* guidance, Reachy captures one frame per checkpoint, a VLM judges whether the person appears confused, and a constrained rule-and-LLM policy chooses to offer a hint, encourage, or add nothing. Both use identical stories, answer keys, and hints. Because the strategies differ in both camera access and when help is offered, we interpret differences as differences between complete strategies, not as the effect of the camera alone.

Our work addressed the following **research questions**:

- **RQ1:** Compared with fixed-point guidance, how does camera-informed guidance change participants' ratings of helpfulness, guidance timing, and the match between the help received and when they wanted it?
- **RQ2:** How do fixed-point and camera-informed guidance differ in participants' final multiple-choice accuracy and initial-to-final answer improvement?

The following code implementation exist here: https://github.com/fanning13/HRI-lab333.git (the study code is in `guidance_study/`). Video of Reachy Mini in the HRI scenario (four clips): https://drive.google.com/drive/folders/1KQmbK1nYUo2-PXA0Jl994M648heiw137. Not all participants consented to video and image sharing, so the clips include only participants who agreed.



Fig. 1. Experimental setup showing a participant interacting with Reachy Mini during the development of the collaborative storytelling system.

2. Background

A. Research theme 1: Recognizing Support Needs

Identifying when users may benefit from assistance is an important challenge in HRI. Li and Ross investigated confusion during situated, task-oriented interactions with a social robot [1], and then examined confusion-inducing interactions and the relationship between self-reported confusion and observable signals including gaze, head pose, facial affect indicators, speech, and silence [2]. These studies show the relevance of multimodal cues, while highlighting that observable behavior is not definitive evidence of an internal mental state. A

participant looking away may be concentrating rather than confused, so using visual observations to inform assistance requires cautious interpretation. The gap we address is that this work studies recognizing confusion, not how a camera-derived cue, treated as uncertain and combined with typed evidence, changes the timing and reception of robot help in a reasoning task. We examine the effect on the overall interaction rather than confusion-detection accuracy.

B. Research theme 2: Generative AI, Multimodal Guidance, and User Experience

Kim, Lee, and Mutlu examined design considerations for LLM-powered HRI [3], and Abbo and Belpaeme demonstrated a vision-enabled dialogue system that brings visual observations into robot conversation [4]. Janssens and colleagues predicted user enjoyment in human–robot dialogue with LLMs [5], emphasizing subjective experience. Zabala and colleagues explored collaborative storytelling with a social robot, generative language models, and visual object recognition, and reported limitations related to turn-based interaction and inference delay [6]. These works motivate using language and visual context in robot responses. The gap we address is the lack of a controlled, end-to-end comparison of always-offered versus selectively offered *identical* hints in structured mystery-solving, evaluated with both perceptions and objective answer outcomes.

3. Robot Control Architecture

Goals. The system goal is to deliver optional guidance reliably while letting the participant retain control of answer choices. The VLM-centered system problem is whether a robot that interprets one camera frame, together with typed evidence, can decide *when* to offer a hint better than a robot that always offers one. This follows from the gaps above: observable cues are only weakly diagnostic of confusion [1], [2], vision can be added to robot dialogue [4], and generative latency disrupts turn-taking [6]. We assess the complete robot behavior with objective measures (final-answer accuracy, initial-to-final improvement, hints offered and delivered, and case completion time), not the quality of model outputs.

Inputs. Participants type an A/B choice, a written explanation, any explicit request for help, revisions, and ratings at a terminal. The system also receives the story context, checkpoint identifier, answer key, and predefined hint. In the camera-informed condition only, Reachy captures one camera frame as soon as the checkpoint question has been spoken. No participant speech is recorded or recognized.

Outputs. Reachy speaks the story narration, checkpoint questions, hint offers, hints, a neutral encouragement line, rating prompts, and the closing through text-to-speech (TTS); the same text is printed in the terminal. The system writes structured CSV logs. The VLM does not generate motor commands or unrestricted dialogue.

Modules. The architecture has six modules (Figure 2).

1) **Story and Checkpoint Controller:** reads `task_materials.json`, presents six story turns per case, manages checkpoints at turns 2, 4, and 6, and retrieves questions, answer keys, scoring criteria, and hints.
2) **Terminal Input and Answer Scoring:** collects initial and final answers, explanations, help requests, and ratings; scores the letter choice against a fixed key and gives a secondary explanation score from deterministic, checkpoint-specific evidence checks.
3) **Frame Acquisition and VLM Interpretation** (camera-informed only): captures and saves one frame per checkpoint and, while the participant types, queries the `gemma3:4b` VLM (local, via Ollama). It returns a cautious description and whether the person appears confused (`yes`, `no`, or `unclear`); the cue is never spoken.
4) **Guidance Decision Policy:** in the fixed-point condition it offers the same hint at every checkpoint without the camera. In the camera-informed condition, rules apply in order: typed help, hint, clue, or stuck delivers the checkpoint hint; typed uncertainty triggers a hint offer; a VLM cue of `yes` triggers a hint offer; a missing frame, unclear cue, or model failure produces the neutral line "Take your time—you're making progress"; otherwise a local `llama3.2` model chooses among `offer_hint`, `neutral_encouragement`, and `no_nudge`.
5) **Robot Speech and Behavior Controller:** synthesizes and caches spoken lines with `gpt-4o-mini-tts` and plays them on Reachy Mini. Voice and movement settings, and the text of accepted hints, are identical across conditions.
6) **Interaction Logger:** records participant, order, case, condition, checkpoint, answers, scores, ratings, VLM and policy fields, hint events, nudge source, timing, and speech fields.

After a participant submits an initial answer, the policy selects and delivers a response; both conditions then ask whether to keep or change the answer, record the final answer, and ask for a confusion rating (1–5) about the moment before Reachy responded. Three post-case ratings follow each case.

Architecture type. The system is a hybrid deliberative-reactive, semi-autonomous architecture. A deterministic controller manages story progression, inputs, guidance rules, and robot responses (deliberative structure), while the VLM and LLM contribute context at checkpoints and may choose only among permitted dialogue actions (reactive component). We chose it because it balances contextual adaptability with experimental control: predefined rules, a fixed set of prompts and hints, and typed participant input keep robot behavior predictable and trials comparable. Reactions occur at checkpoints, not through continuous camera monitoring. The system is semi-autonomous because it selects responses without an operator, while a facilitator sets up equipment and obtains consent and participants decide whether to accept hints and revise answers. The VLM never controls motion.

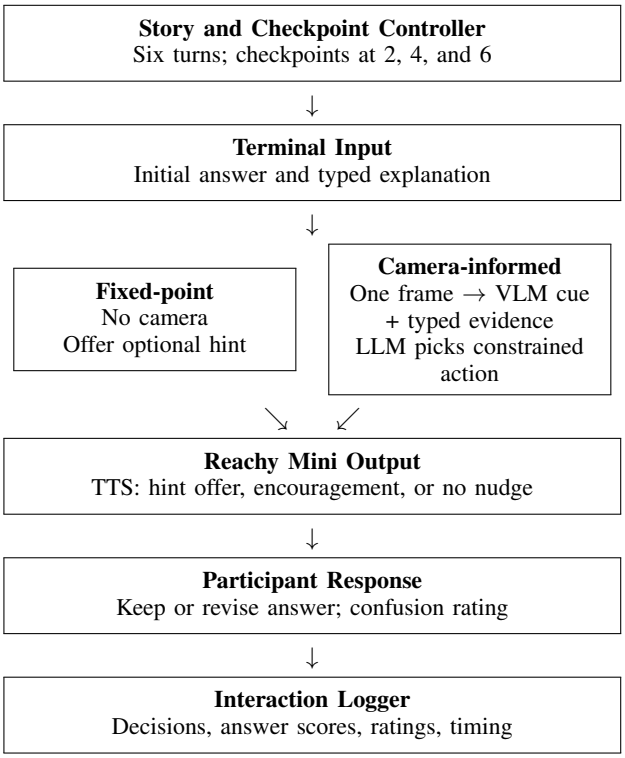


Fig. 2. Robot control architecture comparing fixed-point and camera-informed guidance. Camera acquisition and VLM/LLM inference occur only in the camera-informed condition.

4. Evaluation

A. Experimental Design

The independent variable is guidance strategy with two levels, **fixed-point** and **camera-informed**, described in Section 3. The study is within-participant: every participant completes both conditions, each with a different mystery case. We chose this because the sample is small and between-person differences in reasoning would otherwise be large relative to the effect of guidance. Counterbalancing crosses case, condition, and order in four sequences (Table I); participants P01–P04 were assigned to sequences 1–4 and P05–P08 in the same order.

TABLE I
Counterbalancing sequences.

Sequence	First case	Second case
1	Ashcombe / Fixed	Briarwick / Camera
2	Briarwick / Camera	Ashcombe / Fixed
3	Ashcombe / Camera	Briarwick / Fixed
4	Briarwick / Fixed	Ashcombe / Camera

The dependent variables follow the research questions. For RQ1: post-case ratings (1–5) of helpfulness, timing, and match to wanted help. For RQ2: final-answer accuracy and initial-to-final improvement, plus a secondary keyword-based explanation score. System-level context: hints offered, accepted, and delivered, nudge source, checkpoint confusion rating (1–5), response time, and case completion time. The paired comparison speaks directly to the questions: RQ1 is answered by paired ratings and RQ2 by paired accuracy and improvement.

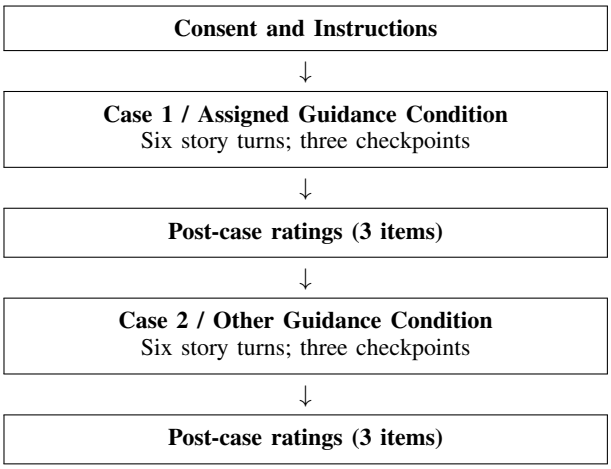


Fig. 3. Within-participant evaluation workflow. Case assignment and guidance order are counterbalanced across four sequences.

B. Study Trials

Eight adult volunteers, all Cornell Tech graduate students who can read short English-language mystery scenarios, took part on October 19, 2026, recruited by convenience sampling. They are identified by pseudonymous IDs (P01–P08), two per sequence. Participation was voluntary; consent to retention and sharing of video and images was separate, and not every participant gave it. Each participant completed two case-level trials with three checkpoints each (six checkpoint observations). In total there are 16 case-level trials and 48 checkpoint observations (24 per condition), plus 48 post-case rating records. Checkpoint observations are nested within participants and are not treated as independent samples.

Development history and conditions. Our original design was a spoken storytelling interaction. We spent roughly three to four days debugging microphone capture on the robot without usable recordings. The Reachy Mini daemon logs show repeated application restarts (the conversation app's background tool manager started again within minutes) and WebSocket errors such as "WebSocket is not connected," followed by disconnections. Because speech input could not be made reliable, on the final day we replaced the design with the typed-input mystery task, wrote the task materials and controller, implemented both conditions, and recruited and ran participants within that day, so there was no pilot phase. The sessions took place in a shared laboratory where other teams were testing audio-enabled models, so background noise distracted from a task that needs quiet reasoning. Some sessions had to be attempted more than once and were handled before analysis.

C. Study Task

Each mystery has six predefined story turns and three checkpoints at turns 2, 4, and 6. Participants examine evidence about a missing valuable object and answer A/B questions about the last confirmed location, logged key access, and whether access alone is proof of theft. In *Ashcombe Star* a silver brooch disappears from a museum display case; in *Briarwick Watch* a pocket watch goes missing from a hotel archive. Both cases have parallel structure, answer keys, explanation criteria, and checkpoint hints.

The facilitator explains the terminal procedure and obtains consent, and Reachy narrates the case with TTS. At each checkpoint the participant chooses an initial answer and types an explanation; the assigned strategy then determines Reachy's response (an optional hint offer, encouragement, or nothing). The participant can accept or decline a hint, keeps or changes the answer, gives a final explanation, and rates how confused they felt just before Reachy responded (1–5). After each case they rate helpfulness, timing, and match (1–5). The procedure repeats for the second case under the other strategy. Explicit help requests receive the same checkpoint hint in both conditions, and an incorrect answer alone is never treated as confusion.

Autonomy. Reachy operates semi-autonomously: it runs the interaction and selects responses without an operator, the human types all input and decides whether to accept hints, and the facilitator handles setup and consent but does not choose dialogue actions. The LLM/VLM pipeline is part of the camera-informed workflow as described in Section 3.

D. Metrics

Data sources and counts. The system logs checkpoint interactions and post-case ratings, collected at inference time only; no model is trained. Each checkpoint record has the participant, order, case, condition, checkpoint, initial and final answers and explanations, correctness, answer change, explanation score, confusion rating, hint offered/accepted/delivered, nudge source, and timing. At each camera-informed checkpoint the program saves one JPEG frame under `guidance_study/frames/`, named by participant, order, checkpoint, and UTC timestamp. Frames are stored separately from the CSV and matched by name because the frame-path column is not populated in the exported logs; they are retained only for consenting participants (the retained set comes from P01–P04) and are not in the public repository. The exported CSVs have 168 rows (48 checkpoint, 48 post-case rating, 48 story, 16 case-start, and 8 session-end rows). They lack populated VLM or reasoning latencies and speech-delivery flags, so we do not report model latency, robot-delivery rate, or VLM sensitivity.

Objective measures. Final-answer accuracy for a condition is

$$A_c = \frac{\text{Correct final answers}}{\text{Valid checkpoints}}, \tag{1}$$

TABLE II
DESCRIPTIVE PAIRED OUTCOMES (8 PARTICIPANTS; MEAN ± SAMPLE SD ACROSS PARTICIPANTS). RATINGS 1–5; ACCURACY AS PERCENT OF THREE CHECKPOINTS PER PARTICIPANT.

Measure	Fixed-point	Camera-informed
Helpfulness (1–5)	2.38 ± 0.74	4.00 ± 0.76
Timing (1–5)	2.25 ± 0.71	3.88 ± 0.99
Help match (1–5)	2.38 ± 0.52	4.13 ± 0.64
Initial accuracy (%)	58.3 ± 15.4	70.8 ± 11.8
Final accuracy (%)	75.0 ± 15.4	95.8 ± 11.8
Improvement (pp)	16.7 ± 25.2	25.0 ± 15.4
Explanation score (%)	66.7 ± 17.8	83.3 ± 17.8
Confusion (1–5)	3.17 ± 0.18	3.17 ± 0.31
Input response (s)	25.78 ± 5.04	26.12 ± 6.51

computed per participant over three checkpoints. Checkpoint improvement is

$$\Delta A = A_{\text{final}} - A_{\text{initial}}, \tag{2}$$

with $+1$ for incorrect-to-correct, 0 for no change, and -1 for correct-to-incorrect, averaged per participant and condition. Accuracy is a task-success rate, a standard task-oriented HRI metric [7]; improvement isolates what the interaction added beyond the participant's first answer; hint offers and deliveries measure intrusiveness; and case completion time (from case introduction through the final explanation at checkpoint three) captures efficiency. Response time is measured from the end of Reachy's question to submission of the initial explanation, so it reflects participant input time, not robot latency. These are computed from logged events, not opinions. The three ratings are subjective and used for RQ1 only.

Analysis. For each participant and condition we compute accuracy, improvement, and each rating, then report means and sample SDs across the eight participants (not across nested checkpoints) with paired camera-minus-fixed differences. We report counts of hint offers, acceptances, nudge types, and revisions, and split ratings by whether a condition came first or second. Confusion ratings are context, not ground-truth labels. The analysis is descriptive, with no hypothesis tests or causal claims; differences are between complete strategies.

5. RESULTS

Results come from the eight sessions described above, run in a noisy shared lab after a last-day redesign; values are descriptive.

Perceptions and accuracy. Camera-informed guidance has the highest mean on all three perception dimensions (Table II). The largest difference is help match ($4.125 - 2.375 = 1.750$); helpfulness and timing each differ by 1.625. Paired camera-minus-fixed differences average 1.625 ± 1.408 (helpfulness), 1.625 ± 1.506 (timing), and 1.750 ± 0.886 (match). Final accuracy is $23/24 = 95.8\%$ for camera-informed versus $18/24 = 75.0\%$ for fixed-point; initial accuracy is $17/24 = 70.8\%$ versus $14/24 = 58.3\%$. The camera condition has six incorrect-to-correct changes and fixed has four; neither

TABLE III
RECORDED EVENT COUNTS (24 CHECKPOINTS PER CONDITION) AND CASE DURATION (8 CASES PER CONDITION). SPEECH-DELIVERY FLAGS WERE NOT POPULATED IN THE EXPORTED LOGS AND ARE NOT REPORTED.

Measure	Fixed-point	Camera-informed
Hint offers	24/24	12/24
Hints marked accepted/delivered in logs	8	9
Incorrect→correct revisions	4	6
Correct→incorrect revisions	0	0
No-nudge responses	0	7
Neutral-encouragement responses	0	5
Case duration (s, mean±SD)	253.1 ± 31.8	256.3 ± 24.0

has correct-to-incorrect changes. Net improvements are $+25.0$ versus $+16.7$ percentage points, but since initial accuracy already differed, the final-accuracy gap is not proof of a better intervention. Explanation scores are 20/24 versus 16/24, and confusion averages 3.17 in both.

Best and worst conditions. Camera-informed is numerically best and fixed-point worst on helpfulness (4.00 vs 2.38), timing (3.88 vs 2.25), help match (4.13 vs 2.38), initial accuracy (70.8% vs 58.3%), final accuracy (95.8% vs 75.0%), improvement (25.0 vs 16.7 percentage points), and explanation score (83.3% vs 66.7%). Confusion is tied (3.17). Fixed-point is marginally best on input-response time (25.78 s vs 26.12 s) and case duration (253.1 s vs 256.3 s), far smaller than the SDs. Fixed-point offered the most hints (24 vs 12). The pattern is consistent: the strategy that offered fewer, selectively triggered hints received higher ratings and had higher accuracy, though the conditions also differ in initial accuracy, case, and period.

Order effects. Mean helpfulness falls from 4.50 for camera-first to 3.50 for camera-second cases, and from 3.00 for fixed-first to 1.75 for fixed-second cases. Timing falls from 4.50 to 3.25 under camera and from 2.75 to 1.75 under fixed. This is consistent with fatigue but is not proof; case differences, learning, and expectations could also explain it. Case duration is about 4.2 minutes in both strategies. Speech success, VLM sensitivity, false-positive rate, and inference latency cannot be computed from the exported fields.

6. DISCUSSION

A. Selective Help and Perceived Appropriateness

The camera-informed policy offered half as many hints (12 versus 24) while receiving higher timing and match ratings, compatible with the idea that always asking becomes repetitive in a reasoning task. For RQ1, camera-informed guidance was rated more helpful, better timed, and better matched in our small sample. Because the conditions differ in several components, we cannot say whether the camera, the VLM, or the controller produced this. Abbo and Belpaeme [4] support the feasibility of vision-contextualized dialogue; our data suggest such context may help decide *when* identical hints are offered.

B. Task Performance, Baseline, and Timing

Part of the final-accuracy gap exists before any hint: initial accuracy already favors the camera condition by 12.5 percentage points. For RQ2, camera-informed guidance has higher final accuracy and larger mean improvement, but initial ability, hint uptake, and per-case difficulty must be considered before crediting adaptivity. Our response-time field measures participants' typing time, not system speed. Zabala *et al.* [6] identify long generative response times as an interaction challenge; separating human deliberation from model delay needs direct measurement of VLM inference and audible robot latency, which our exported logs lack.

C. Repeated Cases, Cue Interpretation, and Design Implications

Ratings were lower in the second case under both strategies, suggesting fatigue or repetitiveness effects in repeated-help designs; a practice round and a break may help. Li and Ross [1], [2] caution against overconfident interpretation of behavioral correlates of confusion. Our design treats the visual output as an uncertain cue, never speaks it, and lets participants decline help. For robot design, our results suggest that a restricted, human-in-the-loop action space with selective offers can be perceived as better timed than always asking, without model-generated speech or motion. Given the sample and confounds, this is a hypothesis for larger studies.

7. FUTURE WORK & LIMITATIONS

Sample, environment, and rushed redesign. Eight participants limit statistical power and generalizability, and the task was redesigned, implemented, and run in one day after microphone problems blocked the original design, so there was no pilot. The shared lab was noisy because other teams were testing audio models, which may have disturbed reasoning and affected both accuracy and ratings; some sessions had to be repeated. A quieter room and a pilot would improve validity and replication.

Confounds, measurement, and evidence. Camera access and assistance policy change together, so vision alone cannot be isolated; a no-camera adaptive condition could do so. Counterbalancing limits but does not remove transfer, annoyance at repeated prompts, and fatigue. Single-frame cues lack temporal context, retrospective confusion and one-item ratings are imperfect, and keyword-based explanation scoring can reject correct paraphrases. Image capture can be checked only through the separately stored frames, which exist only for consenting participants, and the exported fields do not let us verify TTS playback or model timing. Because not every participant consented to image and video sharing, our visual evidence and video cover a subset of sessions. Future work should add calibrated human ratings, temporal cues, synchronized frame, inference, and audio logs, preregistered order analysis, and larger, more diverse samples.

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